

Problem 13.3.1

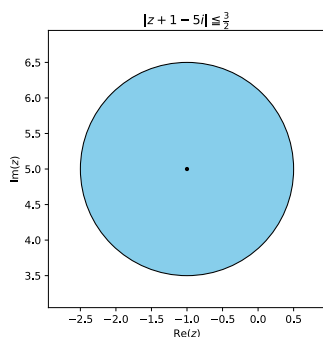
Determine and sketch or graph the sets in the complex plane given by

$$|z + 1 - 5i| \leq \frac{3}{2}$$

Solution

$$|z - (-1 + 5i)| \leq \frac{3}{2}$$

This is a closed disc centered at $(-1 + 5i)$ with radius 1.5



Problem 13.3.3

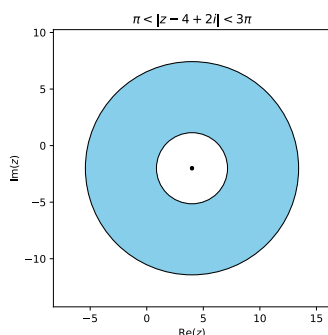
Determine and sketch or graph the sets in the complex plane given by

$$\pi < |z - 4 + 2i| < 3\pi$$

Solution

$$\pi < |z - (4 - 2i)| < 3\pi$$

This is an open disc centered at $(4 - 2i)$ with outer radius 3π , and inner radius π



Problem 13.3.5

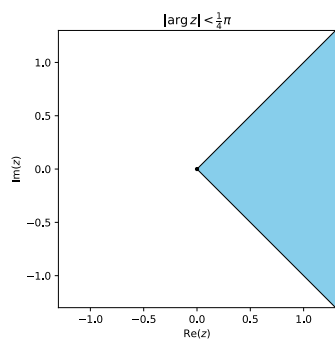
Determine and sketch or graph the sets in the complex plane given by

$$|\arg z| < \frac{1}{4}\pi$$

Solution

$$-\frac{1}{4}\pi < \arg z < \frac{1}{4}\pi$$

This is a wedge from $-\frac{1}{4}\pi$ to $\frac{1}{4}\pi$. There is no bound to the radius of the wedge.



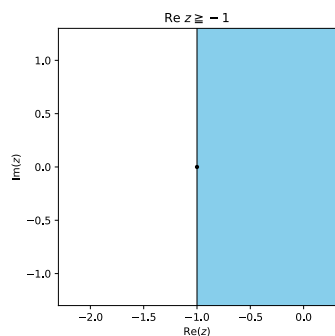
Problem 13.3.7

Determine and sketch or graph the sets in the complex plane given by

$$\operatorname{Re} z \geq -1$$

Solution

This covers the right half of plane bounded at the left by $x = -1$.



Problem 13.3.11

Find $\operatorname{Re} f$, and $\operatorname{Im} f$ and their values at the given point z .

$$f(z) = \frac{1}{1-z} \text{ at } 1-i$$

Solution**Problem 13.3.23**

Find the value of the derivative of

$$\frac{z^3}{(z+1)^3} \text{ at } i$$

Solution**Problem 13.4.3**

Is f analytic?

$$f(z) = e^{-2x}(\cos 2y - i \sin 2y)$$

Solution**Problem 13.4.5**

Is f analytic?

$$f(z) = \operatorname{Re}(z^2) - i \operatorname{Im}(z^2)$$

Solution**Problem 13.4.7**

Is f analytic?

$$f(z) = \frac{i}{z^8}$$

Solution**Problem 13.4.13**

Is the function harmonic? If yes, find a corresponding analytic function $f(z) = u(x, y) + iv(x, y)$.

$$u = xy$$

Solution**Problem 13.4.17**

Is the function harmonic? If yes, find a corresponding analytic function $f(z) = u(x, y) + iv(x, y)$.

$$v = (2x + 1)y$$

Solution**Problem 13.4.23**

Determine a and b so that the given function is harmonic and find a harmonic conjugate.

$$u = \cosh ax \cos y$$

Solution**Problem 13.5.3**

Find e^z in the form $u + iv$ and $|e^z|$ if

$$z = 2\pi i(1 + i)$$

Solution**Problem 13.5.9**

Write in exponential form

$$4 + 3i$$

Solution**Problem 13.5.15**

Fine Re and Im of

$$\exp(z^2)$$

Solution**Problem 13.5.17**

Fine Re and Im of

$$\exp(z^3)$$

Solution**Problem 13.5.19**

Find all solutions and graph some of them in the complex plane.

$$e^z = 1$$

Solution